

Homework: Scope Management Report IT Project Management

Exercises 1-6

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1 Exercise 1: Contract Renegotiation

Problem: Mr. M owes \$1,000 in Year 2 and \$3,000 in Year 5. If they renegotiate to pay the entire amount in Year 3 with an annual interest rate of 6%, what is the required payment?

1.1 Formula

To find the value at a specific point in time ($t = 3$), we move all cash flows to that year:

$$Value_{t=3} = P_2(1+r)^{(3-2)} + \frac{P_5}{(1+r)^{(5-3)}}$$

1.2 Calculation

Given $P_2 = 1,000$, $P_5 = 3,000$, and $r = 0.06$:

- Moving P_2 forward 1 year: $1,000 \times (1.06)^1 = 1,060$ USD.
- Discounting P_5 for 2 years: $3,000 / (1.06)^2 \approx 2,669.99$ USD.

Final Result: Mr. M needs to pay **3,729.99 USD**.

2 Exercise 2: Total Cost of Immediate Purchase

Problem: A computer costs \$2,000 with an additional 12-month installment plan of \$250 per month, based on an interest rate of 18% monthly. What is the immediate full cost?

2.1 Formula

The immediate cost is the sum of the upfront payment and the Present Value (PV) of the installments:

$$PV_{total} = DownPayment + P \times \left[\frac{1 - (1+r)^{-n}}{r} \right]$$

2.2 Calculation

Given $DownPayment = 2,000$, $P = 250$, $r = 0.18$, and $n = 12$:

$$PV_{installments} = 250 \times \left[\frac{1 - (1.18)^{-12}}{0.18} \right] \approx 1,198.31 \text{ USD}$$

Final Result: The total immediate cost is **3,198.31 USD**.

3 Exercise 3: Loan Balance and Refinancing

Problem: A 100M VND loan at 5% interest had 7 annual payments of 10M made. Calculate the remaining balance and the new annual payment for a 5-year payoff.

3.1 Step 1: Calculate Remaining Balance (B_7)

Using the closed-form solution for a loan with constant payments:

$$B_7 = L(1+r)^7 - P \left[\frac{(1+r)^7 - 1}{r} \right]$$

$$B_7 = 100(1.05)^7 - 10 \left[\frac{1.05^7 - 1}{0.05} \right] \approx 140.71 - 81.42 = 59.29 \text{ Million VND.}$$

3.2 Step 2: Calculate New Annual Payment (A)

To pay off 59.29M over 5 years at 5%:

$$A = B_7 \times \left[\frac{r(1+r)^5}{(1+r)^5 - 1} \right] = 59.29 \times \left[\frac{0.05(1.05)^5}{1.05^5 - 1} \right] \approx 13.69 \text{ Million VND.}$$

4 Exercise 4: Future Value of Rental Income

Problem: Rent of \$6,000/year is paid at the end of each year for 5 years and deposited at 6% compound interest. What is the total after 5 years?

4.1 Formula

The future value (FV) of an ordinary annuity:

$$FV = P \times \left[\frac{(1+r)^n - 1}{r} \right]$$

4.2 Calculation

Given $P = 6,000$, $r = 0.06$, and $n = 5$:

$$FV = 6,000 \times \left[\frac{1.06^5 - 1}{0.06} \right] \approx 6,000 \times 5.6371 = 33,822.60 \text{ USD.}$$

5 Exercise 5: NPV and ROI Comparison (Rate: 10%)

We compare Project A and Project B with an initial investment of \$1,000.

5.1 Project A Analysis

Year	0	1	2	3	4	Total
Costs	1,000	0	0	0	0	1,000
Discount Factor	1.00	0.91	0.83	0.75	0.68	
Discounted Costs	1,000	0	0	0	0	1,000
Benefits	0	500	500	500	500	2,000
Discounted Benefits	0	455	415	375	340	1,585
Benefits - Costs	(1,000)	455	415	375	340	585 (NPV)

ROI (Project A): $585/1,000 = 58.5\%$

5.2 Project B Analysis

Year	0	1	2	3	4	Total
Discounted Costs	1,000	0	0	0	0	1,000
Benefits	0	800	800	200	200	2,000
Discounted Benefits	0	728	664	150	136	1,678
Benefits - Costs	(1,000)	728	664	150	136	678 (NPV)

ROI (Project B): $678/1,000 = 67.8\%$

Selection: Project B should be selected due to the higher NPV and ROI.

6 Exercise 6: Advanced Financial Analysis (Rate: 10%)

Analysis of projects with complex multi-year cash flows.

6.1 Project A Analysis

Year	0	1	2	3	4	5	Total
Costs	500	0	100	0	0	0	600
Discount Factor	1.00	0.91	0.83	0.75	0.68	0.62	
Discounted Costs	500.0	0.0	83.0	0.0	0.0	0.0	583.0
Benefits	0	250	350	450	500	-100	1,450
Discounted Benefits	0.0	227.5	290.5	337.5	340.0	-62.0	1,133.5
Benefits - Costs	(500.0)	227.5	207.5	337.5	340.0	(62.0)	550.5 (NPV)

ROI (Project A): $550.5/583.0 = 94.4\%$

6.2 Project B Analysis

Year	0	1	2	3	4	5	Total
Costs	100	0	50	0	0	0	150
Discounted Costs	100.0	0.0	41.5	0.0	0.0	0.0	141.5
Benefits	0	-100	200	200	200	300	800
Discounted Benefits	0.0	-91.0	166.0	150.0	136.0	186.0	547.0
Benefits - Costs	(100.0)	-91.0	124.5	150.0	136.0	186.0	405.5 (NPV)

ROI (Project B): $405.5/141.5 = 286.6\%$